Shiftless Collatz Model and the Ternary Billiards:
A Geometric Interpretation of the 3n+LSB Trajectory

Hiroshi Harada — March 27, 2026

Abstract

This document presents a geometric interpretation of the Shiftless Collatz Model, a rewrite system mathematically equivalent to the classical Collatz iteration but free from information loss.
By mapping the trajectory onto a $\log_3$-lattice plane and interpreting it as a billiard motion, the model reveals hidden physical structure—*warp*, *inertia*, and *target convergence*—within the chaotic dynamics of the $3n + \mathrm{LSB}$ transformation.

Figure 1. Shiftless Collatz Ternary Billiards (Seed *n* = 27).

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